

Time Series Analysis and Seasonal Forecasting in R

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A statistical-computing project covering stationarity, autocorrelation, seasonal differencing, SARIMA modelling, forecasting, and out-of-sample validation.

1. Theoretical Process

The process is $x_t = w_t^2 w_{(t-1)}^2 - 4$, where w_t is Gaussian white noise with mean zero and standard deviation two.

Quantity	Result
$E(w_t^2)$	4
$E(w_t^4)$	48
$E(x_t)$	12
$\text{Var}(x_t)$	2048
$\text{Cov}(x_t, x_{(t-1)})$	512
$\rho(+/-1)$	0.25
$\rho(h), h > 1$	0

2. What the Theoretical Exercise Shows

The process has a constant mean and variance, and its covariance depends only on lag. It is therefore weakly stationary. Although the underlying innovations are independent, adjacent observations share one squared innovation, producing positive lag-one dependence. At lags beyond one, observations share no innovations and covariance falls to zero.

The covariance function measures the magnitude of lagged dependence. The ACF standardizes that dependence, producing a compact memory pattern: correlation equals one at lag zero, 0.25 at lags plus or minus one, and zero thereafter.

The simulation demonstrates the difference between theoretical and sample behaviour. A finite sample will not reproduce the exact theoretical moments, but larger samples should move closer to them.

3. Bridge from Theory to SARIMA

This first exercise prepares the reader to interpret the unemployment analysis. In the theoretical model, the dependence structure is known exactly. In the economic series, it must be inferred from the observed data.

Tool	Role in Script 02
Time plot	Reveals changing level, trend, seasonality, and instability
Mean and variance	Describe the level and dispersion of the series
Covariance	Measures dependence between observations at different lags
ACF	Shows persistence, seasonality, and moving-average structure
PACF	Highlights direct autoregressive relationships by lag
First differencing	Removes persistent changes in level
Seasonal differencing	Removes recurring annual patterns

Tool	Role in Script 02
SARIMA	Models remaining short-run and seasonal dependence
Residual diagnostics	Checks whether systematic dependence remains

The progression is therefore deliberate: Script 01 explains why stationarity and lag dependence matter; Script 02 uses plots, differencing, ACF/PACF diagnostics, model fitting, and residual checks to estimate those structures in a real seasonal series.

4. Unemployment Time-Series Analysis

The applied series contains monthly U.S. unemployment levels from January 1948 through December 1978, expressed in tens of thousands of persons. The analysis applies first and seasonal differencing and fits SARIMA(2,1,0)(0,1,1)[12].

5. Forecasting Workflow and Prediction

Step	Operation
1	Plot monthly unemployment levels, 1948-1978
2	Inspect original ACF and PACF
3	Apply first differencing
4	Apply seasonal differencing at lag 12
5	Fit SARIMA(2,1,0)(0,1,1)[12]
6	Forecast January-December 1979
7	Compare forecast with observed 1979 values

Unemployment stood at approximately 5.896 million persons in December 1978. The model projected a winter rise to 6.950 million in January and 7.040 million in February 1979, a spring decline to 5.729 million in May, a rebound to 6.842 million in June, and stabilization near 6.025 million in December.

The predicted direction was therefore a winter increase, spring decline, mid-year rebound, and year-end stabilization around six million unemployed persons.

6. Monthly Forecast and Validation

Month	Forecast	Observed	Error
Jan 1979	6.950 M	6.600 M	-350 K
Feb 1979	7.040 M	6.650 M	-390 K
Mar 1979	6.719 M	6.320 M	-399 K
Apr 1979	6.042 M	5.720 M	-322 K
May 1979	5.729 M	5.420 M	-309 K
Jun 1979	6.842 M	6.410 M	-432 K
Jul 1979	6.676 M	6.280 M	-396 K
Aug 1979	6.316 M	6.320 M	4 K
Sep 1979	6.153 M	5.980 M	-173 K
Oct 1979	5.899 M	5.960 M	61 K
Nov 1979	6.088 M	5.960 M	-128 K
Dec 1979	6.025 M	6.030 M	5 K

The observed 1979 series followed the same broad seasonal shape. The model generally overpredicted the first seven months, with mean absolute error near 247,500 persons. The August forecast was within roughly 4,000 persons, and the December forecast of approximately 6.025 million was within roughly 2,000 persons of the observed 6.027 million.

Overall, the model captured the seasonal direction and year-end level well while overstating unemployment through much of early 1979.

7. Repository Organization

The repository separates theoretical exercises, applied forecasting, data documentation, generated outputs, validation data, and reporting materials.